

# Privacy-Preserving Machine Learning: Federated Approaches and Challenges

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## Abstract

This paper presents a survey of recent advances in the field. We review key contributions, discuss methodological trends, and identify open challenges. Our analysis covers both foundational work and recent developments, highlighting directions for future research.

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## 1. Introduction

Federated learning enables training across decentralized data without sharing raw samples. The seminal FedAvg algorithm [1] demonstrated communication-efficient optimization by aggregating local model updates. Machine learning models are vulnerable to adversarial perturbations [2], and adversarial training [3] has become a standard defence. Differential privacy techniques [4] provide formal guarantees against privacy leakage during training.

## 2. Related Work

Membership inference attacks [5] reveal that model outputs can leak information about training data, motivating privacy-preserving training methods. Asynchronous federated protocols have been proposed for heterogeneous client availability [6], while differentially private gradient methods have been adapted to tree-based models for healthcare [7]. Secure aggregation protocols [8] protect individual updates from the server. Byzantine robustness has been addressed via gradient clipping strategies [9], and machine unlearning in federated settings [10] allows clients to retract their data contributions.

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